



Published in final edited form as:

Clin Cancer Res. 2025 August 14; 31(16): 3483–3493. doi:10.1158/1078-0432.CCR-24-2525.

Direct comparison of alternative blood-based approaches for early detection and diagnosis of HPV-associated head and neck cancers

Michael E. Bryan, BSc^{1,2,*}, Ling Aye, MD^{1,2,*}, Dipon Das, PhD^{1,2,*}, Shun Hirayama, MD^{1,2,3,*}, Yana Al-Inaya^{1,2}, Julia Mendel, BS^{1,2}, Saskia Naegele, MD^{1,2}, Vasileios Efthymiou, MD^{1,2}, Bayan Alzumaili, MD^{2,4,5}, William C. Faquin, MD, PhD^{2,4,5}, Peter M. Sadow, MD, PhD^{2,4,5}, Derrick Lin, MD^{1,2}, Mark A. Varvares, MD^{1,2}, Allen L. Feng, MD^{1,2}, Daniel G. Deschler, MD^{1,2}, Annie W. Chan, MD^{4,6}, Jonathan Paly, DO^{4,6}, Jong C. Park, MD⁷, Thomas Roberts, MD, MBA⁷, Ross Merkin, MD⁷, Sambit K. Mishra, PhD^{8,9}, Lea Kröller, MSc¹⁰, Birgitta Michels, PhD¹⁰, A. John Iafrate, MD, PhD^{4,5}, Lori J. Wirth, MD⁷, Viktor A. Adalsteinsson¹¹, Mathew Crowson, MD^{1,2}, Tim Waterboer, PhD¹⁰, Lisa Mirabello, PhD⁹, Michael S. Lawrence, PhD^{4,7,11}, Zoe Guan, PhD¹², Adam S. Fisch, MD, PhD^{2,4,5}, Jeremy D. Richmon, MD^{1,2}, Daniel L. Faden, MD^{1,2,4,11,¥}

¹Department of Otolaryngology-Head and Neck Surgery, Harvard Medical School, Boston, Massachusetts

²Massachusetts Eye and Ear, Boston, Massachusetts

³Department of Otorhinolaryngology-Head and Neck Surgery, Wakayama Medical University, Wakayama, Japan

⁴Massachusetts General Hospital, Boston, Massachusetts

⁵Department of Pathology, Massachusetts General Hospital, Boston, Massachusetts

⁶Departments of Radiation Oncology, Massachusetts General Hospital and Harvard Medical School, Boston, Massachusetts

⁷Massachusetts General Hospital Cancer Center, Boston, Massachusetts

⁸Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, Rockville, Maryland

⁹Cancer Genomics Research Laboratory, Frederick National Laboratory for Cancer Research, Frederick, Maryland

¹⁰Infections and Cancer Epidemiology, German Cancer Research Center (DKFZ)

¹¹Broad Institute of MIT and Harvard, Cambridge, Massachusetts

¹²Massachusetts General Hospital Biostatistics, Boston, Massachusetts

¥Corresponding author: Daniel Faden, MD, Department of Otolaryngology-Head and Neck Surgery, Harvard Medical School.

Division of Head and Neck Surgical Oncology, Massachusetts Eye and Ear, 243 Charles St, Boston, MA 02118 (dfaden@mgb.org).

*Equal contribution

Abstract

Purpose: The incidence of human papillomavirus-associated head and neck squamous cell carcinoma (HPV+HNSCC) is rising in the United States. Currently there are no early detection approaches for HPV+HNSCC. Two blood-based analytes for early detection and diagnosis of HPV+HNSCC, circulating tumor HPV DNA (ctHPVDNA) and HPV early protein antibodies (HPV Ab), show promise; yet current approaches lack adequate diagnostic accuracy for broad clinical utility. Further, performance metrics across various assays for detecting these analytes alone, or in combination, have not been compared head-to-head. To address these limitations and knowledge gaps, we developed a multi-feature HPV whole genome sequencing (WGS) liquid biopsy for improved low-level ctHPVDNA detection. We defined the performance characteristics of this WGS-based approach and compared this head-to-head with existing blood-based HPV detection approaches to determine the optimal single or combinatorial biomarker strategy for future prospective study of HPV+HNSCC early detection.

Experimental Design: We tested blood samples from 304 participants: 152 untreated incident HPV+HNSCC patients (77% stage I) and 152 general population control patients. We compared WGS-based ctHPVDNA detection, singleplex ddPCR-based ctHPVDNA detection, multiplexed ddPCR-based ctHPVDNA detection, multiplexed HPV Ab detection, and clinical standard of care tissue biopsy, bench-marked to gold standard HPV+HNSCC tissue diagnosis. We then modeled the operational feasibility of these approaches as screening biomarkers for HPV+HNSCC.

Results: HPV WGS sensitivity and specificity were 98.7% and 98.7%. Singleplex ddPCR sensitivity and specificity were 94.2% and 98.6%. Multiplex ddPCR sensitivity and specificity were 90.6% and 96.3%. HPV Ab sensitivity and specificity were 86.4% and 96.3%. A combinatorial approach using both HPV WGS and HPV Ab yielded a sensitivity and specificity of 87.4% and 98.8%. In a head-to-head comparison, HPV WGS demonstrated significantly improved diagnostic accuracy compared to ddPCR (Youden Index for HPV WGS, 0.99 vs ddPCR, 0.90; $P<0.001$), HPV Ab (HPV WGS, 0.99 vs HPV Ab, 0.83; $P<0.001$), and clinical workup (HPV WGS, 0.99 vs clinical workup 0.82; $P<0.001$), which was maintained when evaluating only early stage disease cases. For men ages 55–74, HPV WGS yielded the lowest number needed to screen (2,903 men) and highest positive predictive value (2.6).

Conclusion: HPV WGS-based ctHPVDNA detection demonstrated the highest sensitivity, specificity, diagnostic accuracy and thus lowest number needed to screen and highest positive predictive value compared to ddPCR-based ctHPVDNA detection, HPV Ab-based detection and combinatorial approaches. These results highlight the promise of HPV WGS liquid biopsy for screening early detection of HPV+HNSCC and the need for modeling and cost-effectiveness studies to evaluate and guide screening implementation.

Introduction

The incidence of human papillomavirus (HPV)-associated head and neck squamous cell carcinomas (HPV+HNSCC) is rising in the United States (US)¹. Strategies for early detection could have meaningful public health benefits by diagnosing patients with subclinical disease, permitting the use of less morbid treatments for earlier stage disease and potentially improving survival as well. HPV+HNSCC can be detected with multiple

blood-based analytes, both at and before clinical diagnosis. The two most extensively studied biomarkers are HPV early protein antibodies (HPV Ab) and circulating tumor HPV DNA (ctHPVDNA).

The first, HPV Ab has been shown to be a promising biomarker of HPV+HNSCC. HPV E6 Ab are present in ~83% of HPV+HNSCC patients at diagnosis, ~5% of HPV-independent HNSCC patients at diagnosis (~95% tumor specificity) and are rare among individuals without cancer in the general population (population specificity ~99%)²⁻⁶. Combinatorial approaches including multiple early antibodies (E1, E2, E5, E6, E7), and bacterially expressed E6 antigen-based approaches show improved sensitivity (~90%) with no change in specificity²⁻⁶. The second, ctHPVDNA, is based on detection of short double-stranded HPV genome fragments in the circulation. ctHPVDNA is detectable in ~89% of HPV+HNSCC patients at diagnosis with the most common detection approach, droplet digital PCR (ddPCR), and has a specificity of ~97% in the general population⁷. While ddPCR ctHPVDNA detection has been shown to outperform standard of care diagnostic approaches⁸, and is being used clinically^{9,10}, these approaches remain too insensitive for broad clinical deployment as stand-alone (tissue-biopsy-free) diagnostic tests^{7,11,12}. Sensitivity has been found to increase with next generation sequencing (NGS)-based approaches (≥91%) while maintaining specificity^{7,13}. Additionally, neither HPV Ab nor existing ctHPVDNA detection assays function as prognostic biomarkers at the time of diagnosis, further limiting their clinical utility. Known genomic prognostic features in HPV+HNSCC include the presence of PIK3CA mutations, viral genome integration events, and specific HPV16 single nucleotide polymorphisms (SNPs)¹⁴⁻¹⁶.

Critically, both HPV Ab and ctHPVDNA are also detectable prior to HPV+HNSCC diagnosis and thus are being studied as cancer screening biomarkers alone, and in combination. At present, there are no screening tests for HPV+HNSCC. HPV Ab can develop up to 28 years prior to cancer diagnosis^{3,5,17,18}. At age 50, HPV16 E6-seropositive men have a 50% cumulative risk of developing HPV+HNSCC and a five-year risk of 7.3%³. However, the long and variable lead time between HPV Ab seroconversion and HPV+HNSCC development, and the non-linear nature of the biomarker, means HPV Abs have limited real-world clinical utility. ctHPVDNA is detectable in some patients prior to clinical cancer diagnosis, but sensitivity is limited with current ddPCR-based approaches^{12,19}.

While HPV E6 Abs and ctHPVDNA are highly promising and biologically plausible diagnostic and screening biomarkers for HPV+HNSCC, significantly more sensitive and comprehensive approaches are needed to generate a single or combinatorial assay with high diagnostic accuracy, prognostic capacity and thus clinical utility for screening and diagnosis. Further, head-to-head comparisons are needed to define operational characteristics of ctHPVDNA and HPV Ab alone, and together, as screening tools.

To address these limitations, we developed a multi-feature HPV whole genome sequencing (WGS) liquid biopsy, termed HPV-DeepSeek, for improved low-level ctHPVDNA detection for use in HPV+HNSCC early detection. Here, we conducted a case-control cohort study to define performance characteristics of HPV-DeepSeek and compare this WGS-based

approach, alone, and in combination, head-to-head with existing blood-based HPV detection approaches (singleplex ddPCR-based ctHPVDNA detection, multiplexed ddPCR-based ctHPVDNA detection, multiplexed HPV Ab detection and clinical standard of care tissue biopsy) benchmarked to gold standard HPV+HNSCC tissue diagnosis. We then modeled the operational feasibility of these approaches with their defined performance metrics as screening biomarkers for HPV+HNSCC. Lastly, we examined if machine learning could integrate blood-based features to predict clinical prognostic features, such as disease stage.

Methods

Study design and enrollment:

All participants provided written informed consent to a protocol approved by the Institutional Review Board at Dana Farber/Harvard Cancer Center. This study was conducted in compliance with the U.S. Common Rule. *Cases:* Patients ≥ 18 years old with a new or suspected diagnosis of HNSCC were prospectively identified upon presentation to the head and neck cancer clinics at Massachusetts Eye and Ear and Massachusetts General Hospital and consecutively enrolled. All patients underwent standard of care diagnostic workup. *Controls:* Controls were enrolled 1:1 to cases, and matched to age and sex. Controls were comprised of non-cancer “healthy” population controls and HPV-independent cancer controls, enrolled in a 2:1 fashion, to create a population mix similar to clinical care. Controls were enrolled from the same geographic regions as cases. All blood samples were assigned a study ID and blinded for further analysis.

Clinical diagnostics:

All cancer cases and cancer control patients underwent standard diagnostic workup including practitioners’ choice tissue biopsy, HPV diagnostics (if indicated) per institutional standards and cross-sectional imaging. All cases suspected of being HPV+HNSCC underwent histomorphological assessment and a combination of p16 immunohistochemistry (IHC), DNA PCR, or RNA chromogenic in situ hybridization (ISH). Cancer control patients who did not undergo clinical HPV testing had HPV testing post-hoc by the same anatomic pathology team to ensure completeness of data for analysis of diagnostic specificity. Diagnostic success of HPV+HNSCC was defined as achieving a histomorphological diagnosis of SCC and correct identification of HPV status (associated or independent) in the first diagnostic attempt (biopsy).

Blood and tissue collection and processing for HPV detection:

5–10 ml of blood was collected from all participants and processed for cell free DNA (cfDNA) as previously described⁸. Blood: cfDNA was extracted from 1–4ml of frozen plasma as previously described for all cases and controls⁸. Tissue: DNA was extracted from formalin-fixed, paraffin-embedded (FFPE) tissue blocks after tumor identification and microdissection and sheared prior to sequencing. 20ng of DNA either total cell free from plasma or total DNA from tissue was used as input for all downstream analyses.

ddPCR:

Singleplex and multiplex ddPCR were performed, as previously described⁸. Multiplexed ddPCR was performed using probe mixing for HPV genotypes 16, 18, 33, 35, and 45 to create five distinct clusters for gating which were validated with dilutional experiments in comparison to singleplex ddPCR. Each genotype was targeted by one probe, which have previously been published, which differs from approaches which multiplex multiple probes per genotype⁸. Thresholds for both singleplex and multiplex detection were preset and unchanged from prior reports⁸.

HPV serology:

Multiplexed Luminex-based HPV serology testing was carried out at the German Cancer Research Center (DKFZ) for HPV 16, 33, 35, and 45 using E1, E2, E6 and E7 with pre-set thresholds, as previously described^{18,20}. Serology was considered positive if above the pre-defined threshold and for the correct clinical genotype. Cross-reactivity (incorrect genotype) was considered negative, including for genotypes not included in the panel.

HPV-DeepSeek:

A custom WGS hybrid capture library was designed for 43 HPV genotypes. Libraries were prepared following the KAPA HyperCap cfDNA Workflow v1.0 (Roche) with custom modifications. The steps include: 1. End repair & A-tailing; 2. KAPA Universal UMI adapter ligation; 3. Post-ligation cleanup; 4. Amplification using KAPA UDI primer mixes; 5. Cleanup using KAPA HyperPure beads; 6. Quantification by Qubit and quality check by TapeStation; 7. Preparation of 4-plex DNA sample library pools; 8. Use of customized KAPA HyperCap target enrichment probes for hybridization and amplification of the target libraries; 9. Cleanup of enriched target libraries using KAPA HyperCapture bead kit; 10. Quantification and quality check by Qubit and TapeStation; 11. Pooling and normalization of libraries; 12. Denaturation; and 13. Paired-end Illumina sequencing to ~500x HPV genome coverage per sample.

HPV-DeepSeek Informatics:

Custom informatic pipelines were designed to process and analyze the sequencing data. The initial steps involved adapter trimming and quality filtering, followed by alignment to a custom reference genome encompassing human hg38 and 64 HPV reference genomes using BWA-MEM. After identifying Unique Molecular Indexes (UMI) and performing deduplication, HPV genotypes were defined by mapping to the HPV reference genome. The number of unique HPV-mapped UMIs in each sample was used to determine ctHPVDNA read counts. For HPV16 samples with > 50% genome coverage, position-wise base counts were obtained, and a consensus sequence was created by selecting the most prevalent nucleotide at each position. These consensus sequences were fed into the multi-alignment tool MUSCLE, along with multiple reference builds for HPV16 sublineages. The resulting alignment was fed into PhyML (run using TOPALi v2.5), assuming a transversion model, allowing for invariant sites and specifying a gamma distribution. The resulting phylogenetic tree enabled sublineage assignment of each sample by nearest distance to a sublineage reference genome. Once aligned to the appropriate reference genome, HPV SNPs were

called using bcftools, filtered based on AF (> 0.5) and read depth (≥ 4). Human single-nucleotide variants (SNVs) in *PIK3CA* were called using MuTect v1 and annotated by SnpEff, with the cut-off value of $AF \geq 0.04$. Mutations calls from plasma samples were validated in matched tumor tissue when available. Viral-human integration events were identified by BWA-MEM. The breakpoints of these events were filtered to ensure chimeric reads were truly composed of a human-genome part and virus-genome part, and at least 4 high-quality reads supporting these events were required. All integration events were manually reviewed to ensure accuracy in IGV. Individual breakpoints were then counted and reported as the number of integration events, per patient. We then evaluated whether breakpoints were clustered within a patient, using the methods described by Symer et al, with the addition of using DBSCAN (<https://scikit-learn.org/1.5/modules/generated/sklearn.cluster.DBSCAN.html>) to perform the clustering²⁸. Integration events in plasma samples were validated in matched tumor tissue when available, using HPV-DeepSeek. Cancer gene lists were obtained from the OncoKBTM Database. The complete list of genes used for annotation was obtained from the GENCODE database. For fragmentomic analyses, fragment length was measured using CollectInsertSizeMetrics (Picard tools) and data was stratified based on alignment to human chromosomes 1–22 or HPV genomes, and patient of origin (HPV+HNSCC, cancer control, or healthy control). Calculations of skewness, kurtosis, and standard deviation were done using SciPy (version 1.11.4) and pandas (version 1.5.4).

HPV-DeepSeek threshold determination:

Optimal thresholds for sensitivity and specificity were established using serial dilutions (1ng – 0.00001ng) of HPV+HNSCC patient cfDNA into a mixture of control patient cfDNA run in triplicate (Supplemental Figure 1, Supplemental Table 1). Cut points for a positive assay were determined to be ≥ 10 HPV reads and $\geq 10\%$ HPV genome coverage and were utilized for all subsequent analyses.

Tissue-based HPV WGS:

DNA from 26 FFPE tissue samples underwent WGS using both HPV-DeepSeek and an amplicon-based HPV WGS developed by Mirabello and colleagues, referred to here as the NCI CHANGeS (Carcinogenic HPV All Next Generation Sequencing) assay and considered the reference gold standard for HPV genome annotation. This assay uses Ion AmpliSeq panels designed to amplify the whole-genomes of the 13 high risk HPV types concurrently. WGS was performed for all of the HR-HPV types concurrently on the Ion PGM platform. Raw sequencing reads were quality-controlled and adaptor trimmed using the Torrent Suite Software (TMAP) and aligned to the HPV reference genomes using the Torrent Mapping Alignment Program. Variant calling was performed using the Torrent Variant Caller (TVC) and SnpEff was used for variant annotation^{21–23}. Three samples were excluded from the study due to low-quality reads. For viral SNP variant calling, AF thresholds were set at >0.5 for both ‘NCI CHANGeS’ and HPV-DeepSeek.

Modeling:

Number needed to screen (NNS) and positive predictive value (PPV) were calculated to determine the diagnostic performance efficiency between ctHPVDNA-based approaches

(ddPCR and WGS) in comparison to HPV Ab, following the method utilized by Lou et al²⁴ and as shown here.

$$PPV = \frac{\text{Sensitivity} \times \text{incidence}}{(\text{Sensitivity} \times \text{incidence}) + ([1 - \text{specificity}] \times [1 - \text{incidence}])}$$

An incidence NNS = $\frac{1}{\text{Sensitivity} \times \text{incidence}}$ rate of 34.9 per 100,000 was used for HPV-associated oropharynx cancer in men aged between 55–74, according to Center for Disease Control statistics (<https://gis.cdc.gov/Cancer/USCS/#/RiskFactors/>). For modeling a two-step approach, performance metrics of HPV Ab were first applied as above (step 1), followed by HPV-DeepSeek to only the population screening positive on step 1 (step 2).

Statistical analysis and hypothesis testing:

The objective of this study was to compare the diagnostic accuracy of ctHPVDNA-based detection approaches (HPV WGS, multiplex ddPCR, singleplex ddPCR), and HPV Ab-based approaches to first attempt clinical biopsy, benchmarked to clinical gold standard diagnosis. Diagnostic accuracy was measured by Youden indexes as described by Chen et al²⁵. We tested the hypothesis that HPV WGS has improved diagnostic accuracy over other ctHPVDNA-based approaches, HPV Ab, combinatorial approaches (HPV WGS and Ab) and clinical standard of care. Statistical significance was assessed using a Z test to compare Youden indices across the same number of cases and controls, with kappa statistics, variances, covariances, confidence intervals, and p-values calculated to determine the significance of differences. Multiplex ddPCR was used as the primary ddPCR approach for comparisons as this is the most commonly used approach clinically^{10,11,26}. Sensitivity for detection in early-stage disease was evaluated by Z test for comparisons of Youden indices across assays, while Fisher's tests were employed for between-group comparisons. Wilcoxon rank sum tests were used for within-assay comparisons. The accuracy of blood-based annotation of known prognostic features, including PIK3CA mutations, integration events, genotype/sub-lineage assignment, and HPV16 high-risk SNPs, were validated using tissue sequencing with HPV-DeepSeek for both human and viral features and with NCI CHANGeS for viral features. Kendall τ coefficient was used to assess the monotonic relationship between ctHPVDNA burden and clinical variables, while Cliff's Delta provided a measure of effect size (Supplemental Table 2). To assess the relationship between ctHPVDNA levels and disease characteristics, we compared ctHPVDNA read counts across different groups within T stage, N stage, overall stage, and integration status. Bootstrap techniques were used to estimate 95% confidence intervals for the differences in median ctHPVDNA read counts between these groups. Spearman correlation coefficient was calculated to assess the relationship between ordinal clinical variables (e.g., T stage, N stage, and overall stage) and the number of unique reads, to enable an understanding of how tumor characteristics correlate. Correlation coefficients (Spearman and Kendall τ) were interpreted as follows: less than ± 0.10 as very weak, ± 0.10 – 0.19 as weak, ± 0.20 – 0.29 as moderate, and strong as ± 0.30 or greater²⁷. Prior to multivariate analysis, a Box-Cox transformation was applied to normalize the outcome variable for further analysis. The Shapiro-Wilk test was then performed on the transformed data to verify statistical assumptions of normality.

Machine learning:

Data pre-processing involved binning categorical staging into grouped categories to create discrete target variables. To ensure consistent scaling across input variables, we normalized features using *StandardScaler* for applicable models. The dataset was then split into training (70%), validation (15%), and test sets (15%) using the *train_test_split* function. The training set was used to train the model, the validation set for hyperparameter tuning, while the withheld test set provided an evaluation of the final model's performance on unseen data. We designed a multi-class machine learning approach to predict T, N, and overall stages. Using scikit-learn's *MultiOutputClassifier*, we trained a single multi-label, multi-class model to predict all three staging components. This framework applied interpretable model architectures, including Logistic Regression, Random Forest, and Gradient Boosting Machine. Model performance was assessed through accuracy, precision, recall, and F1-score for each target variable. We applied a bootstrap method to calculate 95% confidence intervals for these metrics. To evaluate whether our models performed significantly better than naïve guessing, we established a baseline classifier that predicts the most frequent class (or bin) for each target and conducted permutation tests for each target variable (naïve classifier). This statistical test involved randomly shuffling the true labels of the target variable and comparing the model's accuracy against these permuted labels across 1,000 permutations. Analyses were performed using Python version 3.11.4, Jupyter notebook version 6.5.4, and scikit-learn version 1.3.0.

Data Availability:

Sequencing data generated in this study are available in SRA: (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1249105/>) and (<https://www.ncbi.nlm.nih.gov/bioproject/PRJNA1249425/>). Additional raw data such as those used to generate figures are available from the corresponding author upon request.

Results

Cohort demographics and clinical diagnostic workup.

152 HPV+HNSCC patients were enrolled. The mean age was 62 years (range 36–84). The cohort was predominately male (88%; 134/152) with American Joint Committee on Cancer (AJCC) 8th edition stage I (77%; 117/152) oropharyngeal cancer (91%; 138/154) (Table 1). 152 population-level controls were enrolled to match the cases 1:1 with a distribution of healthy controls to HPV-independent HNSCC controls of 2:1. Cancer cases underwent physicians' choice clinical work up which led to fine-needle aspiration (FNA) of a lymph node in 53% of the cohort (81/152 cases) and primary tumor biopsy in 47% (71/152), as first diagnostic attempt. On the first diagnostic attempt, the diagnostic success rate was 68% (55/81) for FNA and 97% (69/71) for tissue biopsy, with an overall diagnostic success rate of 82% (124/152), meaning 18% of the cohort required repeat biopsy (Table 1). p16 IHC was the most common HPV diagnostic test ordered on first diagnostic attempt with a diagnostic success rate of 64% (34/53) with FNA, 99% (68/69) with tissue and 84% (102/122) overall (Supplemental Table 3, 4). A specific HPV genotype was assigned in 50% of cases (76/152).

HPV-DeepSeek diagnostic accuracy and genotyping accuracy

cfDNA from all the cases and controls were run on HPV-DeepSeek (Figure 1A). The mean number of ctHPVDNA reads for HPV+HNSCC cases was 48,997 (range 8–812,564) with a mean genome coverage of 90% (range 9–100%), resulting in a sensitivity of 98.7% and specificity of 98.7% using pre-specified thresholds (Figure 1B and Supplemental Figure 1). Sensitivity remained statistically unchanged when restricted to only oropharynx cancer (sensitivity 98.6%), early stage disease (AJCC 8 stage I sensitivity 98.2%), and when restricting to the very earliest cases (T1 with 0–1 lymph nodes ≤ 3 cm sensitivity 94.1%) (Supplemental Table 5). HPV genotype was assigned in 100% of cases (152/152) by HPV-DeepSeek. Eight genotypes were detected in the cohort, which matched clinical genotyping in 99% of cases (75/76) (Figure 1C, Supplemental Table 6). In the single case of discordant result between clinical tissue-based genotyping and HPV-DeepSeek, clinical PCR detected HPV 16 while two distinct genotypes were detected by HPV-DeepSeek, HPV16 with 4,990 reads and 100% genome coverage and HPV67 with 6,624 reads and 99.2% genome coverage, resulting in HPV67 being assigned as the dominant genotype due to a higher read count. HPV16 was the dominant genotype across the cohort (88%; 134/152) followed by HPV33 (3%; 5/152), 45 (3%; 4/152), 35 (2%; 3/152), 56 (2%; 3/152) and 69, 67, 18 (1 each) (Figure 1C, Supplemental Table 6). HPV16 sublineages were assigned in 98% of HPV16 cases (131/134), with HPV16 A1 being the dominant sublineage (69%; 90/131) (Figure 1C).

Head-to-head comparison of blood- and tissue-based diagnostic approaches

HPV-DeepSeek, HPV Ab, singleplex ddPCR, multiplex ddPCR and first attempt clinical biopsy were compared, benchmarked to clinical gold standard diagnosis (Table 2). Singleplex ddPCR sensitivity was 94.2% (98/104) and specificity was 98.6% (69/70). Multiplex ddPCR sensitivity was 90.6% (96/106) and specificity was 96.3% (77/80). HPV serology sensitivity was 86.4% (121/140) and specificity was 96.3% (4/108) (Figure 1D). A combinatorial approach using both HPV-DeepSeek and HPV Ab, requiring both tests to be positive for a positive test, yielded a sensitivity of 87.4% (90/103) and specificity of 98.8% (80/81). In a head-to-head comparison of >100 overlapping cases for each comparison, HPV-DeepSeek demonstrated significantly improved diagnostic accuracy compared to ddPCR (HPV-DeepSeek Youden Index 0.990 vs multiplexed ddPCR Youden Index 0.904, $P<0.001$), HPV Ab (HPV-DeepSeek Youden Index 0.986 vs. HPV Ab Youden Index 0.827, $P<0.001$) and clinical workup first biopsy ((HPV-DeepSeek Youden Index 0.987 vs. clinical work up Youden Index 0.816, $P<0.001$) (Figure 1E, Supplemental Table 7). HPV-DeepSeek remained more sensitive and with resultant statistically higher diagnostic accuracy for early stage disease (Supplemental Table 8). For example, HPV-DeepSeek sensitivity and diagnostic accuracy for AJCC 8 stage I disease was 98.7% and 0.987 vs. ddPCR, 89.9% and 0.861, $p=0.024$. HPV-DeepSeek sensitivity and diagnostic accuracy for T1 with 0–1 lymph nodes ≤ 3 cm was 100% and 1.0 vs ddPCR 54.5% and 0.508 $p=0.001$). HPV-DeepSeek and ddPCR HPV read counts were highly correlated in positive samples ($R=0.95$, 95% CI 0.91–0.97) with HPV-DeepSeek correctly detecting HPV in samples that were ddPCR-negative due to low reads (Figure 1F).

Blood-based genomic prognostic feature annotation

High-risk SNPs: 23 HPV+HNSCC cases underwent NCI CHANGeS HPV whole genome sequencing assay to generate gold-standard genome annotation from tumor tissue as well as plasma and tumor sequencing with HPV-DeepSeek. Viral genome mutation rates were varied across samples (Figure 2A, Supplemental Figure 2). There was high concordance for positive mutation calls between the assays: HPV-DeepSeek plasma vs. HPV-DeepSeek tumor: 0.902, HPV-DeepSeek plasma vs. NCI CHANGeS: 0.821, HPV-DeepSeek tumor vs. NCI CHANGeS: 0.873 (Figure 2B, Supplemental Figure 2A–D). The concordance for both positive and negative mutation calls was equal to or nearly 100%: HPV-DeepSeek plasma vs. HPV-DeepSeek tumor: 1.000, HPV-DeepSeek plasma vs. NCI CHANGeS: 0.999, HPV-DeepSeek tumor vs. NCI CHANGeS: 0.999 (Supplemental Table 9). Annotation of eight known high-risk HPV16 SNPs demonstrated 100% concordance across all assays (21/21 concordant calls)¹⁶. 16% of HPV16 cases (21/134) contained ≥ 1 high-risk SNP, consistent with prior tissue-based sequencing studies (Figure 2C, D)¹⁶. *PIK3CA*: 9% of cases (14/152) contained high confidence *PIK3CA* mutations (Figure 2E). Of the 14 *PIK3CA* mutations, five were known APOBEC-induced hotspot mutations. *Viral integration events*: 34% (52/152) of cases contained ≥ 1 high confidence integration event (breakpoint). Of 231 total events, 65% (150/231) mapped to human genes. The number of integration events varied widely between patients (mean 4.4, range 1–40). As described previously by Symer *et al*²⁸, breakpoints within individual tumors were frequently clustered (Figure 2F). Breakpoints were unevenly distributed across the viral genome and human genome. Normalized to length, E1 was the viral gene with the most breakpoints, and chr9 was the human chromosome with the most breakpoints (Figure 2G–H, Supplemental Table 10A–B). Breakpoint frequency was significantly higher in cancer-related genes compared to non-cancer genes ($P < 0.001$) (Figure 2I, Supplemental Table 10C). The HPV genome showed no preferential loss in any region, except for polyadenylation sites (Figure 2J).

Blood-based clinical prognostic feature annotation

As reported previously, levels of ctHPVDNA at diagnosis correlated positively with T stage ($P < 0.001$), N stage ($P < 0.001$), and overall stage ($P = 0.05$) (Supplemental Figure 3)^{27,29}. Multivariate regression analysis identified N stage as the strongest predictor of ctHPVDNA levels (Supplemental Table 11). Cases with multiple high-risk features detected from the blood (high-risk SNPs, *PIK3CA* variants, integration events) had a higher overall stage ($P < 0.01$) and higher ctHPVDNA read count ($P < 0.01$) (Supplemental Figure 4). Fragmentomic features were analyzed. ctHPVDNA demonstrated a significantly shorter mean fragment size (148 bp) than total cfDNA from HPV+HNSCC patients (167 bp) and healthy controls (169 bp) ($P < 0.001$) (Figure 3A, B). The mean read length ratio between ctHPVDNA and cfDNA was consistently lower across HPV+HNSCC cases (Supplemental Figure 5). ctHPVDNA also exhibited broader distribution of fragment sizes ($P < 0.001$) and increased skewness ($P < 0.001$) (Figure 3C, D). Next, to evaluate the potential of integrating all HPV-DeepSeek features (Figure 3E) to predict clinical prognostic features such as disease stage from the blood, multiclass classifiers were trained and tested using all blood-based HPV-DeepSeek outputs with an emphasis on predicting nodal stage, a crucial clinical feature in determining treatment approach. Several models, including decision trees, random

forests, support vector machines, and ensemble methods were judged based on their ability to handle both numerical and categorical data, for their interpretability, as well as robustness to overfitting. Logistic Regression demonstrated the best accuracy in the held-out test set, achieving 0.87 (95% CI: 0.74–1.00) for N stage, 0.87 (95% CI: 0.70–1.00) for overall stage and 0.70 (95% CI: 0.52–0.87) for T stage (Supplemental Table 12, Supplemental Figure 6). Classification accuracy was significantly higher than a baseline guessing strategy for T stage ($P=0.020$), N stage ($P=0.001$), and overall stage ($P=0.006$) (Supplemental Table 13).

Modeling HPV+HNSCC early detection

We modeled screening efficiency of HPV-associated oropharynx cancer, the majority subset of HPV+HNSCC in men using diagnostic accuracy of ctHPVDNA based approaches (HPV-DeepSeek, multiplex ddPCR) and HPV Ab to estimate the NNS and PPV. Using an incidence rate of 34.9 per 100,000 for HPV-associated oropharynx cancer in men ages 55–74, HPV-DeepSeek yielded the lowest NNS (2,903 men) and highest PPV (2.6) compared to multiplex ddPCR (3,163 and 0.8), and HPV Ab (3,316 and 0.8) (Figure 4A, B, Supplemental Table 14). We then modeled a two-step screening approach, as has been proposed previously³, in which first a low cost, long lead time risk biomarker was applied (HPV Ab) followed by the highest accuracy diagnostic biomarker (HPV-DeepSeek) in only patients who were HPV Ab positive. Using this two-step approach, NNS in the second step was 96 and PPV was 44.7 (Figure 4C, D, Supplemental Table 15).

Discussion

We conducted a head-to-head comparison of multiple blood-based HPV+HNSCC ctHPVDNA and HPV Ab early detection approaches in a case-control cohort. We found that HPV WGS-based detection alone demonstrated the highest sensitivity, specificity, and diagnostic accuracy compared to ddPCR-based ctHPVDNA detection, HPV Ab-based detection and combinatorial approaches and thus the lowest number needed to screen and highest positive predictive value when screening efficiency was modeled. Integration of blood-based features from HPV WGS detection with machine learning resulted in accurate clinical staging. Taken together, these results highlight the potential utility of an HPV WGS liquid biopsy approach for screening early detection and diagnosis of HPV+HNSCC.

The incidence of HPV+HNSCC is rising in the US and thus strategies for early detection could have meaningful public health benefits¹. Both ctHPVDNA detection and HPV Ab detection have shown promise as potential screening biomarkers for HPV+HNSCC. However, at present, both have critical limitations preventing clinical utility and implementation. Primarily, ctHPVDNA detection approaches lack adequate sensitivity for low-level ctHPVDNA detection, such as that seen in screening^{7,12,19} while HPV Ab have a long and variable (up to >25 years) lead time between seroconversion and HPV+HNSCC development, and are non-linear, leading to limited real-world clinical utility when used alone. Combinatorial ctHPVDNA ddPCR and HPV Ab approaches have been evaluated in small cohorts but have not been found to improve performance at cancer diagnosis³⁰. Such limitations could be addressed through significantly more sensitive ctHPVDNA detection approaches alone, or in combination with HPV Ab. Here, we found in head-to-head

comparison that HPV WGS was significantly more sensitive than ddPCR-based ctHPVDNA detection, and that this sensitivity increase was maintained in the earliest stages of disease, where population-level screening is aimed. Because of the extremely high diagnostic accuracy of HPV WGS, combining this approach with HPV Ab did not lead to statistically improved performance at diagnosis.

Beyond improved diagnostic accuracy, an additional advantage of a multi-feature HPV WGS approach is its potential to also function as a prognostic biomarker. In this study we found HPV-DeepSeek accurately annotated known genomic prognostic features from the blood, including integration events, PIK3CA mutations, viral sublineage and high-risk viral SNPs. This suggests that such an approach could not only serve as a screening biomarker but also as an integrated prognostic biomarker for personalized treatment. We further explored if a completely blood-based approach could be used to not only generate genomic prognostic information but predict clinical stage. Integrating HPV-DeepSeek output with machine learning led to overall stage predication accuracy of 0.87, suggesting that with further refinement, accurate stage prediction from only blood-based features may be feasible. Similar to prior studies, ctHPVDNA detection outperformed standard of care clinical diagnosis⁸. Further, HPV-DeepSeek resulted in an accurate HPV genotype assignment 50% more frequently than standard of care. When taken together, it is feasible to imagine that a fully blood-based approach could be used as a longitudinal integrated diagnostic and prognostic biomarker, functioning as a screening detection biomarker, confirmatory diagnostic biomarker, and prognostic biomarker, without the need for traditional standard of care tissue biopsy. Such an approach would require rigorous prospective study and validation.

The high diagnostic accuracy of HPV WGS in this study, even at the earliest stages of disease, suggests screening early detection may be feasible, with the goal of detecting HPV+HNSCC prior to the onset of regional lymph node metastases— the most common presenting symptom at present. Shifting diagnosis to earlier stages of disease could lead to less treatment morbidity by permitting targeted, single modality treatments (surgery or radiation alone) as opposed to the multi-modality treatments typically used at present time (surgery followed by radiation or radiation with chemotherapy), simultaneously improving quality of life and decreasing time toxicity, patient and payor costs, and potentially mortality as well. Prior studies examining HPV+HNSCC screening utilizing HPV Ab alone have suggested screening paradigms in which men would be screened at 50 years old³. We chose to model men ages 55–75 due to the mean age of HPV+HNSCC increasing across time with the most rapidly increasing demographic being men 65–74^{1,31}. Utilizing a framework described by Lou et al²⁴, we modeled NSS and PPV demonstrating lower NNS and higher PPV for HPV WGS than HPV Ab and ddPCR-based ctHPVDNA detection. To put in these numbers in context, a NNS of 2,903 men and a PPV of 2.6 compare favorably to proposed NNS and PPV values for EBV-associated nasopharyngeal carcinoma screening, a similar oncogenic viral cancer type endemic in southern China, in which circulating tumor EBV DNA screening is being studied. There, NNSs of 5,656–2,262 and PPVs of 1.0–2.4% for incidence rates of 20 and 50/100,000 respectively have been proposed as reasonable²⁴. However, these NSS values are elevated above existing cancer types for which screening is recommended in the US (Breast, Colon, Lung, Cervical). Thus, we then modeled a two-step

screening approach, as has been proposed previously³, in which first a low cost, long lead time risk biomarker was applied (HPV Ab) followed by a highly accurate diagnostic cancer biomarker (HPV WGS) in only patients who were positive on first screen. This yielded a NSS of 96 in the second step, on par or lower than existing US cancer screening approaches. These NNS and PPV estimates based on incidence rates are conservative, as a highly accurate screening approach would be expected to detect not only symptomatic incident cases but asymptomatic cancers as well. High specificity is critical for cancer screening to minimize harm to benefit ratio by avoiding unnecessary clinical interventions and their resultant psychosocial and cost impacts from false positives. Individual test specificity of HPV WGS and HPV WGS + HPV Ab was adequately high for use in high incidence populations but would likely not be sufficient for low incidence population use, such as the general population (specificity 98.7–98.8%). Future studies are needed to define, evaluate and model alternative assay cut points which favor specificity (such as sensitivity at 99.5% specificity), as would be needed for population level screening and in settings with ultra-low tumor burden (asymptomatic cases years prior to clinical detection), as well as two-step screening approaches which require both tests to be positive before a more intensive intervention and workup is undertaken. Additionally, further modeling is needed to develop post-blood-based detection clinical work-up algorithms to ensure benefit over harm in concert with cost considerations. Lastly, the high NNS, even with an ultrasensitive test, raises questions regarding the epidemiologic practicality of screening for HPV+HNSCC and suggests early detection may need to focus on specific high risk groups where the incidence of HPV+HNSCC is higher.

Multiple limitations should be noted. Cases and controls were both recruited from hospital clinics. Thus, controls may not reflect the general population (selection bias), although controls recruited from hospital clinics are likely to have increased cancer risk, leading to a bias favoring decreased specificity. Specificity estimates may change when applied in population-based cohorts, which will be a necessary next step. Cases were individuals who were either symptomatic from their cancer or had an incidentally found lesion on cross-sectional imaging or physical exam. Thus, these cases may differ in tumor size (larger) and stage (higher) from a screening setting, leading to over-estimation of sensitivity. Taken together, because this is a EDRN phase 2 biomarker study and not a phase 3 or 4 study, meaning the cases and controls are not from population based studies, the screening modeling results should be interpreted with caution and may not be generalizable to the population³². While the comparison assays used in this study have performance metrics similar to published literature, performance of alternative ddPCR and HPV Ab assays may exceed these, which could theoretically alter findings of the study. For example, multi-probe ddPCR assays may have improved sensitivity compared to single probe ddPCR assays, such as those used here³². Other groups have examined HPV WGS vs ddPCR in cervical cancer, finding improved sensitivity at the time of diagnosis³³. However, they also found no improvement in sensitivity in the minimal residual disease setting after treatment with chemoradiotherapy. The HPV Ab assays used here only target 4 HPV genotypes, inherently decreasing the sensitivity. Lastly, integration events detected in the plasma were validated in tissue by the same methodology, however, additional validation with gold standard long-read WGS of the tumor tissue will be required for additional external validation. The number

of integration events detected in plasma was lower than expected from the tissue-based literature, suggesting we may be under-detecting integration events.

In summary, we conducted the first head-to-head direct comparison of blood-based HPV+HNSCC ctHPVDNA and HPV Ab early detection approaches finding that HPV WGS-based ctHPVDNA detection alone demonstrated the highest sensitivity, specificity, and diagnostic accuracy and thus the lowest number needed to screen and highest positive predictive value when screening efficiency was modeled. These results highlight the promise of HPV WGS liquid biopsy for screening early detection of HPV+HNSCC and the need for modeling and cost-effectiveness studies to evaluate and guide HPV+HNSCC screening.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

The methodology for statistical analysis was facilitated using the Shared Health Research Information Network (SHRINE) with support from Harvard Catalyst | The Harvard Clinical and Translational Science Center (National Center for Research Resources and the National Center for Advancing Translational Sciences, National Institutes of Health Award UL1TR002541 and financial contributions from Harvard University and its affiliated academic healthcare centers. The content is solely the responsibility of the authors and does not necessarily represent the official views of Harvard Catalyst, Harvard University and its affiliated academic healthcare centers, or the National Institutes of Health.

Funding Statement

Funding for this work came from NIH/NIDCR R03DE030550 (Daniel Faden).

Conflicts of Interest

Daniel L. Faden receives salary support from NIH/NIDCR K23 DE029811, NIH/NIDCR R03 DE030550 and NIH/NCI R21 CA267152. Daniel L. Faden has received research funding or in-kind funding from Bristol-Myers Squibb, Calico, Predicine, BostonGene, Neogenomics and Haystack (Quest). He has received consulting fees from Merck, Noetic, Chrysalis Biomedical Advisors, Arcadia, and Focus. None of these sources relate to the work in this manuscript. TW serves on advisory boards for Merck (MSD) Sharp & Dohme. The remaining authors have declared no conflicts of interest.

References

1. Tota JE et al. Evolution of the Oropharynx Cancer Epidemic in the United States: Moderation of Increasing Incidence in Younger Individuals and Shift in the Burden to Older Individuals. *J Clin Oncol* 37, 1538–1546 (2019). [PubMed: 31026209]
2. Hibbert J, Halec G, Baaken D, Waterboer T & Brenner N Sensitivity and Specificity of Human Papillomavirus (HPV) 16 Early Antigen Serology for HPV-Driven Oropharyngeal Cancer: A Systematic Literature Review and Meta-Analysis. *Cancers (Basel)* 13, 3010 (2021). [PubMed: 34208476]
3. Robbins HA et al. Absolute Risk of Oropharyngeal Cancer After an HPV16-E6 Serology Test and Potential Implications for Screening: Results From the Human Papillomavirus Cancer Cohort Consortium. *Journal of Clinical Oncology* 40, 3613–3622 (2022). [PubMed: 35700419]
4. Brenner N et al. Characterization of human papillomavirus (HPV) 16 E6 seropositive individuals without HPV-associated malignancies after 10 years of follow-up in the UK Biobank. *EBioMedicine* 62, 103123 (2020). [PubMed: 33248371]
5. Kreimer AR et al. Timing of HPV16-E6 antibody seroconversion before OPSCC: findings from the HPV3 consortium. *Annals of Oncology* 30, 1335–1343 (2019). [PubMed: 31185496]

6. Lang Kuhs KA et al. Human Papillomavirus 16 E6 Antibodies in Individuals without Diagnosed Cancer: A Pooled Analysis. *Cancer Epidemiology, Biomarkers & Prevention* 24, 683–689 (2015).
7. Naegele S, Ruiz-Torres DA, Zhao Y, Goss D & Faden DL Comparing the Diagnostic Performance of Quantitative PCR, Digital Droplet PCR, and Next-Generation Sequencing Liquid Biopsies for Human Papillomavirus–Associated Cancers. *The Journal of Molecular Diagnostics* 26, 179–190 (2024). [PubMed: 38103593]
8. Siravegna G et al. Cell-Free HPV DNA Provides an Accurate and Rapid Diagnosis of HPV-Associated Head and Neck Cancer. *Clinical Cancer Research* 28, 719–727 (2022). [PubMed: 34857594]
9. Lang Kuhs KA, Brenner JC, Holsinger FC & Rettig EM Circulating Tumor HPV DNA for Surveillance of HPV-Positive Oropharyngeal Squamous Cell Carcinoma. *JAMA Oncol* 9, 1716 (2023). [PubMed: 37824111]
10. Berger BM et al. Detection of Occult Recurrence Using Circulating Tumor Tissue Modified Viral HPV DNA among Patients Treated for HPV-Driven Oropharyngeal Carcinoma. *Clinical Cancer Research* 28, 4292–4301 (2022). [PubMed: 35576437]
11. Hanna GJ et al. Negative Predictive Value of Circulating Tumor Tissue Modified Viral (TTMV)-HPV DNA for HPV-driven Oropharyngeal Cancer Surveillance. *Clinical Cancer Research* 29, 4306–4313 (2023). [PubMed: 37566241]
12. Rettig EM et al. Relationship of HPV16 E6 seropositivity with circulating tumor tissue modified HPV16 DNA before head and neck cancer diagnosis. *Oral Oncol* 141, 106417 (2023). [PubMed: 37148655]
13. Leung E et al. HPV Sequencing Facilitates Ultrasensitive Detection of HPV Circulating Tumor DNA. *Clinical Cancer Research* 27, 5857–5868 (2021). [PubMed: 34580115]
14. Beaty BT et al. PIK3CA Mutation in HPV-Associated OPSCC Patients Receiving Deintensified Chemoradiation. *JNCI: Journal of the National Cancer Institute* 112, 855–858 (2020). [PubMed: 31747025]
15. Koneva LA et al. HPV Integration in HNSCC Correlates with Survival Outcomes, Immune Response Signatures, and Candidate Drivers. *Mol Cancer Res* 16, 90–102 (2018). [PubMed: 28928286]
16. Lang Kuhs KA et al. Genetic variation within the human papillomavirus type 16 genome is associated with oropharyngeal cancer prognosis. *Annals of Oncology* 33, 638–648 (2022). [PubMed: 35306154]
17. Kreimer AR et al. Kinetics of the Human Papillomavirus Type 16 E6 Antibody Response Prior to Oropharyngeal Cancer. *JNCI: Journal of the National Cancer Institute* 109, (2017).
18. Kreimer AR et al. Evaluation of Human Papillomavirus Antibodies and Risk of Subsequent Head and Neck Cancer. *Journal of Clinical Oncology* 31, 2708–2715 (2013). [PubMed: 23775966]
19. Rettig EM et al. Detection of circulating tumor human papillomavirus DNA before diagnosis of HPV-positive head and neck cancer. *Int J Cancer* 151, 1081–1085 (2022). [PubMed: 35262203]
20. Waterboer T et al. Multiplex human papillomavirus serology based on in situ-purified glutathione s-transferase fusion proteins. *Clin Chem* 51, 1845–53 (2005). [PubMed: 16099939]
21. Pinheiro M et al. Association of HPV35 with cervical carcinogenesis among women of African ancestry: Evidence of viral-host interaction with implications for disease intervention. *Int J Cancer* 147, 2677–2686 (2020). [PubMed: 32363580]
22. Pinheiro M et al. Phylogenomic Analysis of Human Papillomavirus Type 31 and Cervical Carcinogenesis: A Study of 2093 Viral Genomes. *Viruses* 13, (2021).
23. Cullen M et al. Deep sequencing of HPV16 genomes: A new high-throughput tool for exploring the carcinogenicity and natural history of HPV16 infection. *Papillomavirus Res* 1, 3–11 (2015). [PubMed: 26645052]
24. Lou P-J et al. Performance and Operational Feasibility of Epstein-Barr Virus–Based Screening for Detection of Nasopharyngeal Carcinoma: Direct Comparison of Two Alternative Approaches. *Journal of Clinical Oncology* 41, 4257–4266 (2023). [PubMed: 37478397]
25. Chen F, Xue Y, Tan MT & Chen P Efficient statistical tests to compare Youden index: accounting for contingency correlation. *Stat Med* 34, 1560–76 (2015). [PubMed: 25640747]

26. Gunning A et al. Analytical Validation of NavDx, a cfDNA-Based Fragmentomic Profiling Assay for HPV-Driven Cancers. *Diagnostics* 13, 725 (2023). [PubMed: 36832208]
27. Rettig EM et al. Association of Pretreatment Circulating Tumor Tissue–Modified Viral HPV DNA With Clinicopathologic Factors in HPV-Positive Oropharyngeal Cancer. *JAMA Otolaryngology–Head & Neck Surgery* 148, 1120 (2022). [PubMed: 36301568]
28. Symer DE et al. Diverse tumorigenic consequences of human papillomavirus integration in primary oropharyngeal cancers. *Genome Res* 32, 55–70 (2022). [PubMed: 34903527]
29. Kentnowski M et al. Determinants of the level of circulating-tumor HPV16 DNA in patients with HPV-associated oropharyngeal cancer at the time of diagnosis. *Sci Rep* 13, 21226 (2023). [PubMed: 38040848]
30. Lewis JS et al. Assessing the feasibility of a multimodal liquid biopsy for the diagnosis of HPV-associated oropharyngeal squamous cell carcinoma. *Am J Clin Pathol* 161, 570–578 (2024). [PubMed: 38349613]
31. Windon MJ et al. Increasing prevalence of human papillomavirus–positive oropharyngeal cancers among older adults. *Cancer* 124, 2993–2999 (2018). [PubMed: 29710393]
32. Pepe MS et al. Phases of Biomarker Development for Early Detection of Cancer. *JNCI Journal of the National Cancer Institute* 93, 1054–1061 (2001). [PubMed: 11459866]
33. Bhambhani C et al. Development of a high-performance multi-probe droplet digital PCR assay for high-sensitivity detection of human papillomavirus circulating tumor DNA from plasma. *Oral Oncol* 143, 106436 (2023). [PubMed: 37269557]
34. Han K et al. Clinical Validation of Human Papilloma Virus Circulating Tumor DNA for Early Detection of Residual Disease After Chemoradiation in Cervical Cancer. *Journal of Clinical Oncology* 42, 431–440 (2024). [PubMed: 37972346]

Translational evidence

The incidence of HPV-associated head and neck cancer (HPV+HNSCC) is rising. Early detection of HPV+HNSCC could have meaningful public health benefits. Two blood-based analytes for early detection and diagnosis of HPV+HNSCC, circulating tumor HPV DNA (ctHPVDNA) and HPV antibodies (HPV Ab) show promise; yet current approaches lack adequate diagnostic accuracy for broad clinical utility. Further, performance metrics across various assays have not been compared head-to-head. We developed a multi-feature HPV whole genome sequencing (WGS) liquid biopsy for improved ctHPVDNA detection. We defined the performance characteristics of this WGS-based approach and compared it with existing blood-based HPV detection approaches to determine the optimal biomarker strategy for HPV+HNSCC early detection. We found that HPV WGS demonstrated the highest sensitivity, specificity, and diagnostic accuracy and thus the lowest number needed to screen and highest positive predictive value when screening efficiency was modeled. These results highlight the promise of HPV WGS liquid biopsy for early detection of HPV+HNSCC.

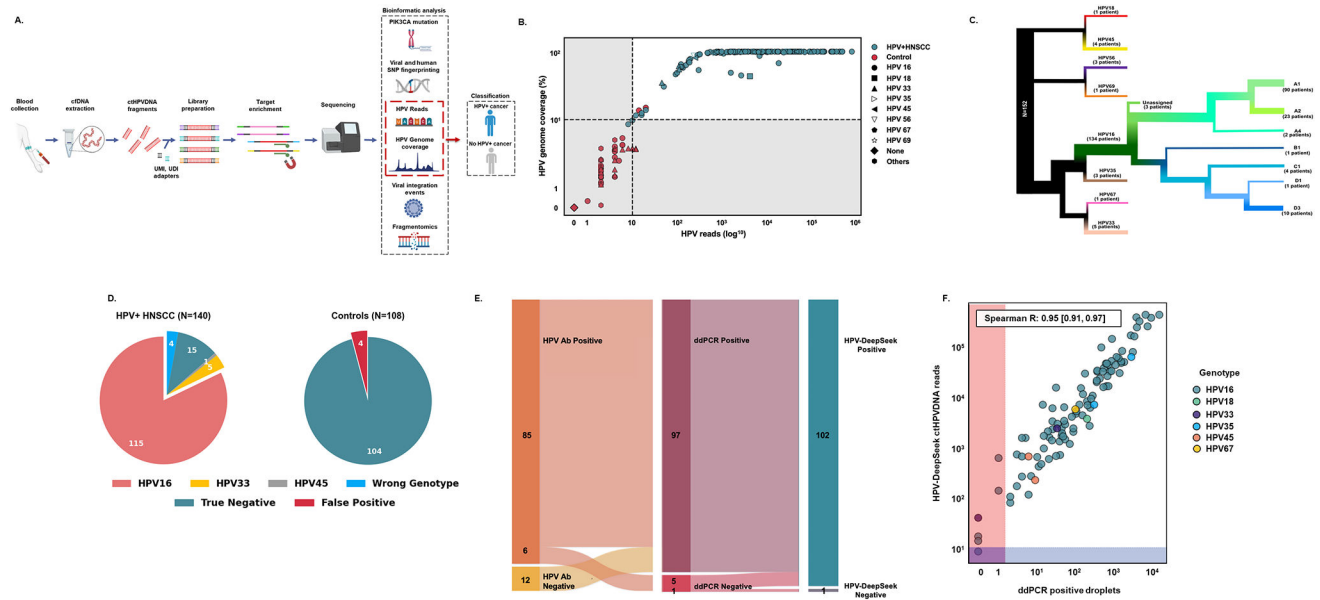


Figure 1. Comparison of blood-based ctHPVDNA and HPV Antibody detection approaches.

A. Schematic representation of HPV-DeepSeek workflow demonstrating two classification approaches: 1. Read/coverage-based classification and 2. Machine learning-based classification. **B.** HPV-DeepSeek diagnostic performance in 152 HPV+HNSCC cases and 152 population-level controls using read/coverage classification. HPV+HNSCC cases are green, controls are red. HPV genotypes are represented in different shapes. The 10 read, 10% genome coverage threshold is highlighted with dotted lines. **C.** Schematic phylogenetic tree of the genotypic diversity among 152 HPV+ HNSCC cases detected by HPV-DeepSeek. Each branch represents a unique HPV genotype, with branch width proportional to the number of cases harboring that genotype. Sublineages are represented in the same format at the end of the HPV16 branch. **D.** Pie chart showing the serology testing of HPV+HNSCC cases (left) and controls (right) with the color corresponding to each category. **E.** Sankey plot showing a head-to-head comparison between three different diagnostic approaches in a cohort of 103 patients. The leftmost flow represents HPV Ab, the middle flow represents ddPCR, and the rightmost flow represents HPV-DeepSeek. Flow width is proportionate to the number of patients. **F.** Scatter plot shows the Spearman correlation between ddPCR positive droplets (x-axis, log scale) and HPV-DeepSeek reads (y-axis, log scale) for 103 HPV+ HNSCC cases analyzed by both assays. Circles in the white region represent samples positive in both assays. Samples in red region are negative by ddPCR but positive by HPV-DeepSeek. Samples in violet region are negative in both assays. Shaded regions approximate the thresholds for HPV-DeepSeek positivity (≥ 10 reads, red) and ddPCR positivity (≥ 2 positive droplets, blue).

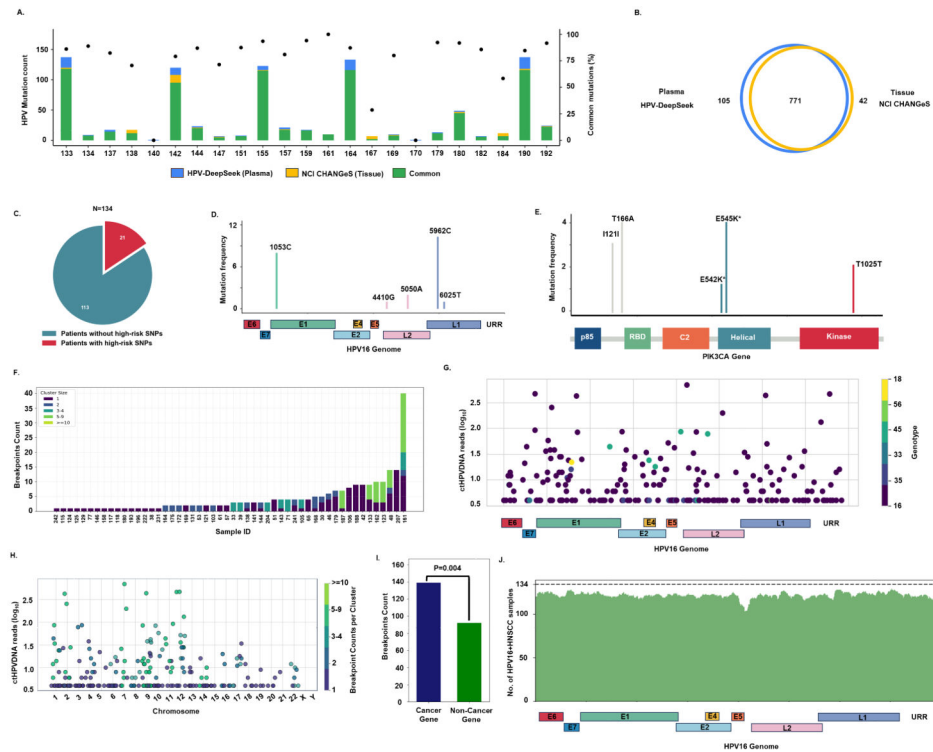


Figure 2. Blood-based genomic prognostic feature annotation.

A. Bar graph representing the total number of HPV mutation counts for an individual sample analyzed by both HPV-DeepSeek and NCI CHANGeS. Green represents the mutations detected in common between the two assays, blue represents mutations observed only in HPV-DeepSeek (plasma) and yellow only in NCI CHANGeS. Black dots denote the percentage of common mutation in each sample. **B.** Venn diagram showing the relationship between HPV-DeepSeek (plasma) and NCI CHANGeS for the total number of HPV mutations found in the 23 samples run on both assays. **C.** Pie chart showing the distribution of high-risk SNPs in 134 patients with HPV16. **D.** Schematic lollipop plot displays the individual high-risk SNPs identified across the HPV16 cases. Lollipop height is number of patients with each mutations. **E.** Schematic lollipop plot represents the different *PIK3CA* gene mutations identified across 152 HPV+HNSCC cases. Lollipop height is the frequency of each given mutation. Asterisk denotes hotspot APOBEC mutations. **F.** Breakpoint counts (y-axis), ordered by frequency in individual HPV+HNSCCs cases (x-axis). Color gradient denotes the breakpoint cluster size. **G.** Breakpoints (circles=231) identified in 52/152 HPV+HNSCC cases uniquely mapped to HPV genotypes (color coded). Within the viral genome, there was a statistically significant difference ($P < 0.05$) between the expected and observed number of breakpoints for E1, E2, E4, L1, L2 and URR. **H.** Breakpoints clustered within 500-kb windows (upstream and downstream) uniquely mapped to the human genome (x-axis, $n = 231$); y-axis denotes breakpoint counts. Color denotes the breakpoint cluster size. For the human genome, there was a statistically significant difference ($P < 0.05$) between the expected and observed number of breakpoints for chromosome 9, 12, X, and Y. **I.** Histogram showing the frequency of breakpoints in cancer vs. non-cancer genes showing a statistically increased number of mutations in cancer genes ($p = 0.004$). **J.** Coverage plot

showing the alignment of 134 HPV16 HNSCC cases to identify preferential genome region loss. The x-axis represents the HPV16 genome while the y-axis represents the number of cases that had reads spanning each genome coordinate.

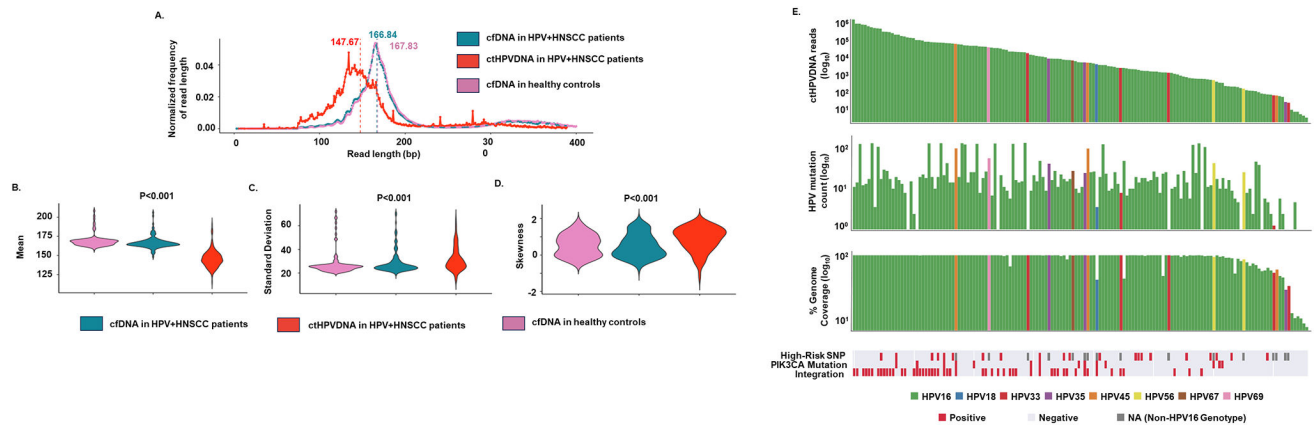


Figure 3. HPV-DeepSeek multi-feature annotation.

A. Comparison of the mean fragment length of 152 HPV+ HNSCC cases from cHPVDNA (red) and non-HPV cfDNA (turquoise), and in 100 healthy controls from cfDNA (pink). The x-axis represents the length of cfDNA reads in base pair (bp), while the y-axis indicates the normalized frequency of read length. **B-D.** Violin plot showing differences in mean (**B**), standard deviation (**C**) and skewness (**D**) of average read lengths (p < 0.001 for all comparisons). **E.** Bar plots showing total number of reads, total genome coverage, and HPV mutation count in 152 HPV+HNSCC cases. The mosaic plot in the lower panel marks the cases that have high-risk SNPs, *PIK3CA* mutation, and HPV integration events. HPV genotypes are color-coded.

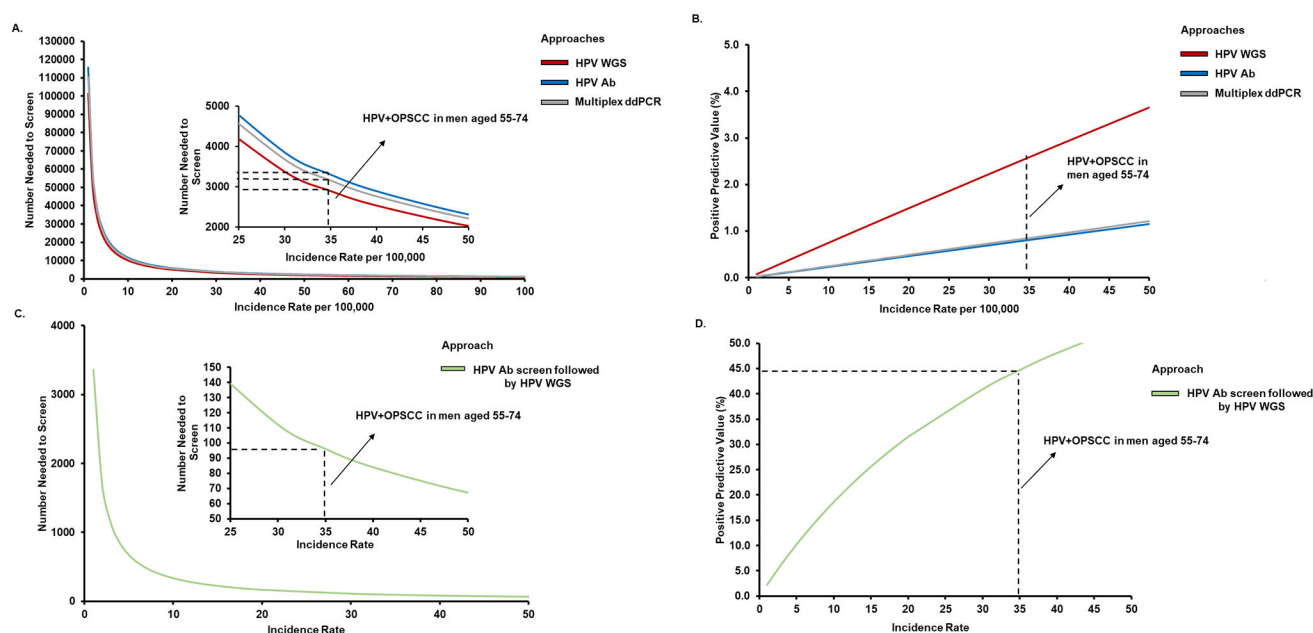


Figure 4 . Diagnostic efficiency estimates of ctHPVDNA and HPV antibody detection approaches.

A. NNS estimates of a single-step screening approach based on sensitivity for HPV-DeepSeek, multiplex ddPCR and HPV Ab for HPV-associated oropharynx cancer (HPV+OPSCC) in men aged 55–74 based on disease incidence. **B.** PPV estimates of a single-step screening approach based on sensitivity and specificity of the three approaches. **C.** NNS estimates of a two-step screening approach with HPV Ab followed HPV-DeepSeek only in the population of HPV Ab screen positive men. **D.** PPV estimates of a two-step screening approach.

Table 1. Patient demographics.

Clinical and demographic characteristics of HPV+HNSCC cohort. Diagnostic success determined by 1. Positive HPV testing and 2. Histopathology for squamous cell carcinoma. Diagnostic success rate is the positive diagnostic success results on first attempt tissue biopsy over total number of patients who had that biopsy type.

Characteristic	HPV+HNSCC (n=152)	Controls (n=152)
Age at blood collection, mean (range)	62 (36–84)	62 (36–84)
Sex		
Male	134	134
Female	18	18
AJCC 8 cancer stage at diagnosis		
I	117	
II	23	
III	7	
IV	5	
Viral genotype per clinical workup		
HPV16	29	
HPV16/18	98	
Other high-risk and Unknown	25	
Primary site		
Oral cavity	3	
Tonsil	65	
Base of tongue	73	
Nasopharynx	2	
Nasal cavity and Paranasal sinuses	3	
Unknown primary	4	
Overlapping	2	
Biopsy performed, n (diagnostic success (n), rate (%))		
Fine-needle aspiration	81 (55, 68%)	
Tissue biopsy	71 (69, 97%)	

Table 2. Blood-based approaches for early detection and diagnosis of HPV-associated head and neck cancers.

Assay performance matrix of multiplex ddPCR (186 cases & controls), singleplex ddPCR (174 cases & controls), HPV Ab (248 cases & controls), HPV WGS (304 cases & controls), and a combination of HPV WGS and HPV Ab, (184 cases & controls). In the combinatorial approach, a sample is considered positive only if both HPV WGS and HPV Ab tests are positive. Assays were evaluated based on four metrics: sensitivity, specificity, positive predictive value, and negative predictive value. Youden Index was calculated for each assay, a composite measure of sensitivity and specificity. HPV WGS had the highest diagnostic accuracy.

Assay		Sensitivity	Specificity	PPV	NPV	Youden Index
ddPCR	Multiplex	90.6 (96/106)	96.3 (77/80)	97.0 (96/99)	88.5 (77/87)	0.868
	Singleplex	94.2 (98/104)	98.6 (69/70) *	99.0 (98/99) *	92.0 (69/75) *	0.929
HPV Ab		86.4 (121/140)	96.3 (104/108)	96.8 (121/125)	84.6 (104/123)	0.827
HPV-DeepSeek		98.7 (150/152)	98.7 (150/152)	98.7 (150/152)	98.7 (150/152)	0.974
HPV-DeepSeek + HPV Ab		87.4 (90/103)	98.8 (80/81)	98.9 (90/91)	86.0 (80/93)	0.862

* Historical data².